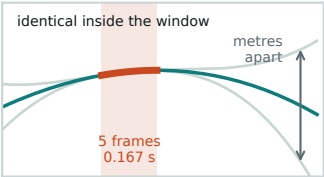


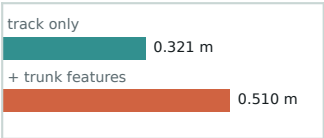
How a height is recovered, and what never sees a pixel

a. Why five frames are not enough



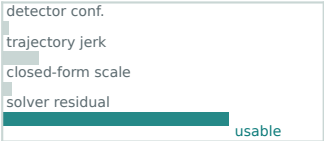
Over 0.167 s the arc is straight, so its height is unobservable. A stroke is long enough to bend.

b. Adding pixels makes it worse



The 64-d trunk feature at the detected point downs the 14-d track it was meant to help.

c. Who knows when the answer is wrong



Nothing the model reports predicts its own error (rank corr. ≤ 0.10). The solver measures its fit after the fact; the gate reads that.

Broadcast frames

5 consecutive frames, 512 x 288, RGB + background difference

TrackNetV5-Lite trunk, fine-tuned

heatmap head kept and supervised $\rightarrow$ soft-argmax (u, v)
F1 95.58 against the released 96.77: the cost is recall, not precision

One stroke of detections

the whole flight, 27 frames at the median, plus the venue camera (four court corners and a focal length, calibrated once)

Height head - one number per frame

6-layer transformer, 192-d, over the stroke
reads the track and the camera. It does not read the image.

Horizontal position - not predicted

$(X, Y) = xy_from_uvz(G, col3, (u, v), Z)$, closed form
exact to 0.031 m on the labels; regressing it costs 0.152 vs 0.113 m

Optional refinement, gated

paper 1's constrained solver, started from OUR estimate
accepted only where its own fit residual is ≤ 20 px (80% of strokes)

3D trajectory in court metres

0.351 m median against paper 1, 6.9% of frames within 5 cm
no camera is estimated by the trajectory head at inference

Height error reaches 3D error through the closed form with a measured gain of 4.1x:
one centimetre of height is four centimetres of position.